EXCEL STATISTICS

GENERAL ARRANGEMENT AND COMPLETE PARTS LIST

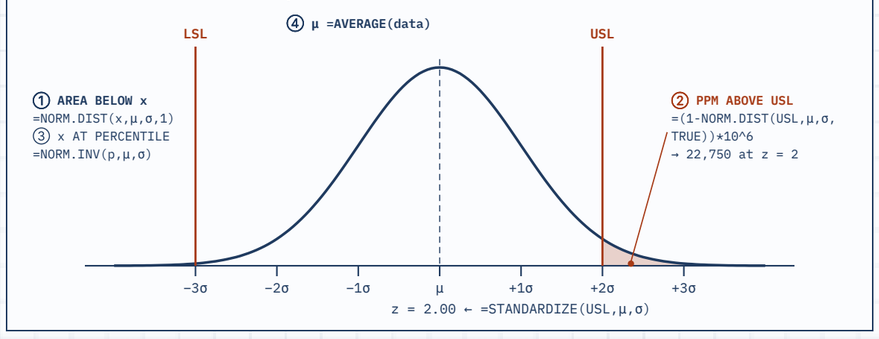
All 111 statistical functions, 21 helpers, 8 What-if tools, 19 ToolPak tools, 39 superseded names and 12 gaps, itemised like a drawing. Item numbers match the lesson's map stops.

GENERAL NOTES

1. Functions per Microsoft 365 Excel. Item 11 names still calculate; do not specify on new work.
2. Use .S functions for samples. STDEV.S is overall σ: Ppk, not Cpk.
3. p-values two-tailed unless noted. Z.TEST returns the upper one-tail p.
4. QUARTILE.EXC matches Minitab quartiles.
5. Parts list 10 requires the Analysis ToolPak add-in.
6. No native normality test: see item 12.01.

DRAWING NO.	REV	SCALE	SHEET	DRAWN	WEB
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VIEW A · THE NORMAL CURVE, DIMENSIONED IN EXCEL



PARTS LIST 01 · DESCRIPTIVE STATISTICS — QTY 36

- 01.01 **AVERAGE**(number1, [number2], ...) — Default centre of a data set
- 01.02 **AVERAGEA**(value1, [value2], ...) — Only when logical values must count
- 01.03 **AVERAGEIF**(range, criteria, [average_range]) — Mean for one shift, grade or line
- 01.04 **AVERAGEIFS**(average_range, criteria_range1, criteria1, ...) — Mean by heat and orientation together
- 01.05 **MEDIAN**(number1, [number2], ...) — Skewed data such as delays and cycle times
- 01.06 **MODE.SNGL**(number1, [number2], ...) — Discrete readings with repeats
- 01.07 **MODE.MULT**(number1, [number2], ...) — Data with two or more peaks
- 01.08 **GEOMEAN**(number1, [number2], ...) — Growth rates, ratios, log-normal data
- 01.09 **HARMEAN**(number1, [number2], ...) — Averaging rates such as units per hour
- 01.10 **TRIMMEAN**(array, percent) — A robust mean when outliers are present
- 01.11 **STDEV.S**(number1, [number2], ...) — Almost every quality calculation on sample data
- 01.12 **STDEV.P**(number1, [number2], ...) — Only when you hold the whole population
- 01.13 **STDEVA**(value1, [value2], ...) — Rarely; text becomes 0 and distorts results
- 01.14 **STDEVPA**(value1, [value2], ...) — Rarely; same caution as STDEVA
- 01.15 **VAR.S**(number1, [number2], ...) — ANOVA, F-tests and pooled variance
- 01.16 **VAR.P**(number1, [number2], ...) — Complete populations only
- 01.17 **VARA**(value1, [value2], ...) — Rarely
- 01.18 **VARPA**(value1, [value2], ...) — Rarely
- 01.19 **AVEDEV**(number1, [number2], ...) — A spread summary that resists outliers
- 01.20 **DEVSQ**(number1, [number2], ...) — Building ANOVA sums of squares by hand
- 01.21 **SKEW**(number1, [number2], ...) — Quick normality and symmetry screen
- 01.22 **SKEW.P**(number1, [number2], ...) — Skewness of a complete population
- 01.23 **KURT**(number1, [number2], ...) — Screening for heavy or light tails
- 01.24 **MAX**(number1, [number2], ...) — Range and worst-case checks
- 01.25 **MAXA**(value1, [value2], ...) — Rarely
- 01.26 **MAXIFS**(max_range, criteria_range1, criteria1, ...) — Worst reading per grade or line
- 01.27 **MIN**(number1, [number2], ...) — Range and minimum-spec checks
- 01.28 **MINA**(value1, [value2], ...) — Rarely
- 01.29 **MINIFS**(min_range, criteria_range1, criteria1, ...) — Lowest toughness per heat
- 01.30 **LARGE**(array, k) — Top-k values and order statistics
- 01.31 **SMALL**(array, k) — Bottom-k values and order statistics
- 01.32 **COUNT**(value1, [value2], ...) — The sample size for every other formula
- 01.33 **COUNTA**(value1, [value2], ...) — Checking for text entries in numeric columns
- 01.34 **COUNTBLANK**(range) — Missing-data audit
- 01.35 **COUNTIF**(range, criteria) — Defects beyond a spec limit
- 01.36 **COUNTIFS**(criteria_range1, criteria1, ...) — Contingency tables and DPU by category

PARTS LIST 02 · PROBABILITY DISTRIBUTIONS — QTY 41

- 02.01 **NORM.DIST**(x, mean, standard_dev, cumulative) — Share of output below a limit, ppm out of spec
- 02.02 **NORM.INV**(probability, mean, standard_dev) — Percentile limits; simulation with RAND()
- 02.03 **NORM.S.DIST**(z, cumulative) — p-values for z statistics
- 02.04 **NORM.S.INV**(probability) — Critical z, sample size, sigma level
- 02.05 **GAUSS**(z) — Quick area from the centre
- 02.06 **PHI**(x) — Plotting the bell curve
- 02.07 **T.DIST**(x, deg_freedom, cumulative) — Lower-tailed t tests
- 02.08 **T.DIST.RT**(x, deg_freedom) — Upper-tailed t tests
- 02.09 **T.DIST.2T**(x, deg_freedom) — Two-sided p-value for a t statistic
- 02.10 **T.INV**(probability, deg_freedom) — One-sided critical values; negative for small p
- 02.11 **T.INV.2T**(probability, deg_freedom) — Confidence intervals and two-sided tests
- 02.12 **CHISQ.DIST**(x, deg_freedom, cumulative) — Lower tail of a variance test
- 02.13 **CHISQ.DIST.RT**(x, deg_freedom) — p-value for chi-square tests with any df
- 02.14 **CHISQ.INV**(probability, deg_freedom) — Upper limit of a variance confidence interval
- 02.15 **CHISQ.INV.RT**(probability, deg_freedom) — Chi-square table lookups; variance intervals
- 02.16 **F.DIST**(x, deg_freedom1, deg_freedom2, cumulative) — Lower tail of a variance-ratio test
- 02.17 **F.DIST.RT**(x, deg_freedom1, deg_freedom2) — ANOVA and regression p-values
- 02.18 **F.INV**(probability, deg_freedom1, deg_freedom2) — Lower limit of a variance-ratio interval
- 02.19 **F.INV.RT**(probability, deg_freedom1, deg_freedom2) — F tables for ANOVA
- 02.20 **BINOM.DIST**(number_s, trials, probability_s, cumulative) — Defectives in a sample, pass or fail data
- 02.21 **BINOM.DIST.RANGE**(trials, probability_s, number_s, [number_s2]) — Acceptance-sampling ranges in one step
- 02.22 **BINOM.INV**(trials, probability_s, alpha) — Acceptance numbers and exact critical counts
- 02.23 **NEGBINOM.DIST**(number_f, number_s, probability_s, cumulative) — Inspections until a set number of finds
- 02.24 **HYPGEOM.DIST**(sample_s, number_sample, population_s, number_pop, cumulative) — Sampling small lots without replacement
- 02.25 **POISSON.DIST**(x, mean, cumulative) — Defects per unit, c- and u-chart probabilities
- 02.26 **EXPON.DIST**(x, lambda, cumulative) — Time between failures, constant failure rate
- 02.27 **WEIBULL.DIST**(x, alpha, beta, cumulative) — Life data and wear-out reliability
- 02.28 **LOGNORM.DIST**(x, mean, standard_dev, cumulative) — Repair times, particle sizes, skewed data
- 02.29 **LOGNORM.INV**(probability, mean, standard_dev) — Percentile limits for skewed data
- 02.30 **GAMMA.DIST**(x, alpha, beta, cumulative) — Waiting times for several events
- 02.31 **GAMMA.INV**(probability, alpha, beta) — Percentiles of gamma-shaped data
- 02.32 **GAMMA**(number) — Building other distributions by hand
- 02.33 **GAMMALN**(x) — Large factorials without overflow
- 02.34 **GAMMALN.PRECISE**(x) — Same use, newer algorithm
- 02.35 **BETA.DIST**(x, alpha, beta, cumulative, [A], [B]) — PERT task times, proportions
- 02.36 **BETA.INV**(probability, alpha, beta, [A], [B]) — Exact binomial confidence limits
- 02.37 **PERMUT**(number, number_chosen) — Counting sequences and orders
- 02.38 **PERMUTATIONA**(number, number_chosen) — Codes and passwords style counting
- 02.39 **PROB**(x_range, prob_range, [lower_limit], [upper_limit]) — Custom discrete distributions
- 02.40 **COMBIN**(number, number_chosen) HELPER — Counting samples where order does not matter
- 02.41 **COMBINA**(number, number_chosen) HELPER — Counting with replacement

PARTS LIST 03 · STATISTICAL INFERENCE — QTY 2

- 03.01 **CONFIDENCE.T**(alpha, standard_dev, size) — Confidence interval for a mean, sigma unknown
- 03.02 **CONFIDENCE.NORM**(alpha, standard_dev, size) — Confidence interval for a mean, sigma known

PARTS LIST 04 · HYPOTHESIS TESTS — QTY 4

- 04.01 **T.TEST**(array1, array2, tails, type) — Comparing two means
- 04.02 **Z.TEST**(array, x, [sigma]) — Mean against a target when sigma is known
- 04.03 **F.TEST**(array1, array2) — Checking variances before choosing T.TEST type
- 04.04 **CHISQ.TEST**(actual_range, expected_range) — Counts by category; contingency tables

PARTS LIST 05 · CORRELATION — QTY 7

- 05.01 **CORREL**(array1, array2) — Strength and direction of a straight-line link
- 05.02 **PEARSON**(array1, array2) — Legacy sheets; identical result
- 05.03 **RSQ**(known_y's, known_x's) — Share of variation explained by one x
- 05.04 **COVARIANCE.S**(array1, array2) — Building r by hand on sample data
- 05.05 **COVARIANCE.P**(array1, array2) — Matches the ToolPak Covariance tool
- 05.06 **FISHER**(x) — Confidence interval on r; comparing two r values
- 05.07 **FISHERINV**(y) — Second half of the r interval

PARTS LIST 06 · REGRESSION — QTY 7

- 06.01 **LINEST**(known_y's, [known_x's], [const], [stats]) — Simple or multiple regression with inference
- 06.02 **SLOPE**(known_y's, known_x's) — Change in y per unit of x
- 06.03 **INTERCEPT**(known_y's, known_x's) — y when x is zero
- 06.04 **STEYX**(known_y's, known_x's) — Width of prediction intervals
- 06.05 **TREND**(known_y's, [known_x's], [new_x's], [const]) — Fitted values, residuals, predictions
- 06.06 **LOGEST**(known_y's, [known_x's], [const], [stats]) — Growth or decay curves
- 06.07 **GROWTH**(known_y's, [known_x's], [new_x's], [const]) — Predictions from an exponential model

PARTS LIST 07 · FORECASTING — QTY 6

- 07.01 **FORECAST.LINEAR**(x, known_y's, known_x's) — Straight-line trend forecasts
- 07.02 **FORECAST**(x, known_y's, known_x's) — Older workbooks only
- 07.03 **FORECAST.ETS**(target_date, values, timeline, [seasonality], [data_completion], [aggregation]) — Time series with trend and seasonality
- 07.04 **FORECAST.ETS.CONFINT**(target_date, values, timeline, [confidence_level], [seasonality], [data_completion], [aggregation]) — Error bands around ETS forecasts
- 07.05 **FORECAST.ETS.SEASONALITY**(values, timeline, [data_completion], [aggregation]) — Confirming the cycle before trusting a forecast
- 07.06 **FORECAST.ETS.STAT**(values, timeline, statistic_type, [seasonality], [data_completion], [aggregation]) — Judging forecast accuracy

PARTS LIST 08 · EXPLORATORY DATA ANALYSIS — QTY 18

- 08.01 **FREQUENCY**(data_array, bins_array) — Histograms and goodness-of-fit counts
- 08.02 **QUARTILE.EXC**(array, quart) — Box plots that match Minitab
- 08.03 **QUARTILE.INC**(array, quart) — Matching older Excel workbooks
- 08.04 **PERCENTILE.EXC**(array, k) — P5 or P95 limits that match Minitab
- 08.05 **PERCENTILE.INC**(array, k) — Percentiles that include 0 and 1
- 08.06 **PERCENTRANK.EXC**(array, x, [significance]) — Where one value sits in the data
- 08.07 **PERCENTRANK.INC**(array, x, [significance]) — Same, inclusive method
- 08.08 **RANK.AVG**(number, ref, [order]) — Normal probability plots and rank tests
- 08.09 **RANK.EQ**(number, ref, [order]) — League tables and Pareto ordering
- 08.10 **STANDARDIZE**(x, mean, standard_dev) — Flagging points beyond three standard deviations
- 08.11 **GROUPBY**(row_fields, values, function, [field_headers], ...) HELPER — Stratify by heat, shift, line or supplier
- 08.12 **PIVOTBY**(row_fields, col_fields, values, function, ...) HELPER — Side-by-side comparisons such as LPA vs TWA
- 08.13 **PERCENTOF**(data_subset, data_all) HELPER — Pareto-style contributions
- 08.14 **FILTER**(array, include, [if_empty]) HELPER — Slicing a subgroup without copying
- 08.15 **UNIQUE**(array, [by_col], [exactly_once]) HELPER — Listing categories before grouping
- 08.16 **SORT**(array, [sort_index], [sort_order], [by_col]) HELPER — Ordering data for plots and ranks
- 08.17 **SUBTOTAL**(function_num, ref1, [ref2], ...) HELPER — Summaries that follow a filter
- 08.18 **AGGREGATE**(function_num, options, ref1, [ref2], ...) HELPER — Robust summaries on messy data

PARTS LIST 09 · WHAT-IF & OPTIMISATION — QTY 18

- 09.01 **Goal Seek** · Data › What-If Analysis › Goal Seek — Which sigma gives Cpk 1.33?
- 09.02 **Data Table (one variable)** · Data › What-If Analysis › Data Table (column input cell) — Sensitivity of cost or yield to one input
- 09.03 **Data Table (two variables)** · Data › What-If Analysis › Data Table (row and column input cells) — Cpk map across mean and sigma
- 09.04 **Scenario Manager** · Data › What-If Analysis › Scenario Manager — Best, likely and worst cases
- 09.05 **Solver (Simplex LP)** · Data › Solver › Select a Solving Method: Simplex LP — Blends, product mix, scheduling
- 09.06 **Solver (GRG Nonlinear)** · Data › Solver › Select a Solving Method: GRG Nonlinear — Response optimization, curve fitting
- 09.07 **Solver (Evolutionary)** · Data › Solver › Select a Solving Method: Evolutionary — Models with IF, LOOKUP or rounding
- 09.08 **Forecast Sheet** · Data › Forecast › Forecast Sheet — One-click seasonal forecast
- 09.09 **RAND()** HELPER — Driving simulations through .INV functions
- 09.10 **RANDARRAY**([rows], [columns], [min], [max], [integer]) HELPER — Thousands of draws in one formula
- 09.11 **RANDBETWEEN**(bottom, top) HELPER — Random sampling of rows
- 09.12 **SEQUENCE**(rows, [columns], [start], [step]) HELPER — Input grids for sensitivity tables
- 09.13 **LET**(name1, name_value1, calculation_or_name2, ...) HELPER — Readable one-cell simulations
- 09.14 **LAMBDA**([parameter1, ...], calculation) HELPER — Custom test statistics and models
- 09.15 **MAP**(array1, lambda_or_array) HELPER — Formula-based data tables that recalculate fast
- 09.16 **SUMPRODUCT**(array1, [array2], ...) HELPER — Objective and constraint cells for Solver
- 09.17 **MMULT**(array1, array2) HELPER — Regression by matrices; linear models
- 09.18 **MINVERSE**(array) HELPER — Normal equations for regression by hand

PARTS LIST 10 · ANALYSIS TOOLPAK · ALL ON DATA › DATA ANALYSIS — QTY 19

- 10.01 **Anova: Single Factor** — Three or more group means
- 10.02 **Anova: Two-Factor With Replication** — Two factors with repeats
- 10.03 **Anova: Two-Factor Without Replication** — Randomized block designs
- 10.04 **t-Test: Paired Two Sample for Means** — Before and after on the same units
- 10.05 **t-Test: Two-Sample Assuming Equal Variances** — Independent groups, equal spread
- 10.06 **t-Test: Two-Sample Assuming Unequal Variances** — Independent groups, unequal spread
- 10.07 **z-Test: Two Sample for Means** — Two means with known variances
- 10.08 **F-Test Two-Sample for Variances** — Comparing two variances
- 10.09 **Descriptive Statistics** — One-click summary
- 10.10 **Regression** — Full regression output
- 10.11 **Correlation** — Several variables at once
- 10.12 **Covariance** — Several variables at once
- 10.13 **Histogram** — Quick frequency table
- 10.14 **Rank and Percentile** — Relative standing table
- 10.15 **Random Number Generation** — Static simulated data
- 10.16 **Sampling** — Drawing an audit sample
- 10.17 **Moving Average** — Trend smoothing
- 10.18 **Exponential Smoothing** — Level-only forecasts
- 10.19 **Fourier Analysis** — Vibration and periodic signals

PARTS LIST 11 · LEGACY NAMES — QTY 39

- 11.01 ~~BETADIST~~ → BETA.DIST
- 11.02 ~~BETAINV~~ → BETA.INV
- 11.03 ~~BINOMDIST~~ → BINOM.DIST
- 11.04 ~~CHIDIST~~ → CHISQ.DIST.RT
- 11.05 ~~CHIINV~~ → CHISQ.INV.RT
- 11.06 ~~CHITEST~~ → CHISQ.TEST
- 11.07 ~~CONFIDENCE~~ → CONFIDENCE.NORM
- 11.08 ~~COVAR~~ → COVARIANCE.P
- 11.09 ~~CRITBINOM~~ → BINOM.INV
- 11.10 ~~EXPONDIST~~ → EXPON.DIST
- 11.11 ~~FDIST~~ → F.DIST.RT
- 11.12 ~~FINV~~ → F.INV.RT
- 11.13 ~~FORECAST~~ → FORECAST.LINEAR
- 11.14 ~~FTEST~~ → F.TEST
- 11.15 ~~GAMMADIST~~ → GAMMA.DIST
- 11.16 ~~GAMMAINV~~ → GAMMA.INV
- 11.17 ~~HYPGEOMDIST~~ → HYPGEOM.DIST
- 11.18 ~~LOGINV~~ → LOGNORM.INV
- 11.19 ~~LOGNORMDIST~~ → LOGNORM.DIST
- 11.20 ~~MODE~~ → MODE.SNGL
- 11.21 ~~NEGBINOMDIST~~ → NEGBINOM.DIST
- 11.22 ~~NORMDIST~~ → NORM.DIST
- 11.23 ~~NORMINV~~ → NORM.INV
- 11.24 ~~NORMSDIST~~ → NORM.S.DIST
- 11.25 ~~NORMSINV~~ → NORM.S.INV
- 11.26 ~~PERCENTILE~~ → PERCENTILE.INC
- 11.27 ~~PERCENTRANK~~ → PERCENTRANK.INC
- 11.28 ~~POISSON~~ → POISSON.DIST
- 11.29 ~~QUARTILE~~ → QUARTILE.INC
- 11.30 ~~RANK~~ → RANK.EQ
- 11.31 ~~STDEV~~ → STDEV.S
- 11.32 ~~STDEVP~~ → STDEV.P
- 11.33 ~~TDIST~~ → T.DIST.RT or T.DIST.2T
- 11.34 ~~TINV~~ → T.INV.2T
- 11.35 ~~TTEST~~ → T.TEST
- 11.36 ~~VAR~~ → VAR.S
- 11.37 ~~VARP~~ → VAR.P
- 11.38 ~~WEIBULL~~ → WEIBULL.DIST
- 11.39 ~~ZTEST~~ → Z.TEST

PARTS LIST 12 · BEYOND EXCEL — QTY 12

- 12.01 **Normality test (Anderson-Darling, Shapiro-Wilk)** — Excel workaround: Normal probability plot with NORM.S.INV and RANK.AVG; check SKEW and KURT
- 12.02 **One- and two-proportion tests** — Excel workaround: z statistic with NORM.S.DIST; exact test with BINOM.DIST
- 12.03 **Nonparametric tests (sign, Wilcoxon, Mann-Whitney, Kruskal-Wallis)** — Excel workaround: Sign test with BINOM.DIST; rank sums with RANK.AVG
- 12.04 **Equal variances, three or more groups (Levene, Bartlett)** — Excel workaround: One-way ANOVA on absolute deviations from each group median
- 12.05 **Post-hoc comparisons (Tukey)** — Excel workaround: Fisher LSD with T.INV.2T; no studentized range in Excel
- 12.06 **Unbalanced or multi-factor ANOVA and DOE analysis** — Excel workaround: LINEST with coded columns for two-level factorials
- 12.07 **Power and sample size** — Excel workaround: Sample-size formula with NORM.S.INV
- 12.08 **Process capability** — Excel workaround: Cpk from AVERAGE and STDEV.S, or R-bar over d2
- 12.09 **Control charts** — Excel workaround: Limits from AVERAGE and constants, drawn as a line chart
- 12.10 **Gage R&R** — Excel workaround: ANOVA method built from DEVSQ
- 12.11 **Weibull fitting** — Excel workaround: LINEST on the linearized Weibull plot
- 12.12 **Logistic or Poisson regression** — Excel workaround: Solver maximizing the log-likelihood
- 12.13 **PY**(python_code, return_type) HELPER — scipy.stats and statsmodels for tests Excel lacks

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